MTH622 MID TERM SOLVED PAPER BY VURANK

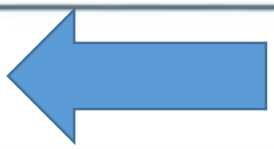
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Question # 6 of 10 (Start time: 03:32:38 PM, 13 June 2011) Total Marks: 1

Greens theorem applies only to _____ vector fields

Select the correct option

o	one dimensional
o	two dimensional
o	three dimensional
o	None of these



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Which of the following theorem convert line integral to surface integral?

Select the correct option

O	Stoke's theorem only
O	Gauss divergence and Stoke's theorem
O	Green's theorem only
O	Stoke's and Green's theorem

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The rate of change of tangential velocity of an object traveling in a circular orbit or path is known as

Select the correct option

O	Normal acceleration
O	Tangential acceleration
O	Rectangular acceleration
O	None of these

Question # 5 of 10 (Start time: 03:45:51 PM, 13 June 2019) Total Marks: 1

Applications of Green's theorem are meant to be in

Select the correct option

- O One dimensional
- O Two dimensional
- O Three dimensional
- O Four dimensional

The normal component of acceleration is zero when motion of particle is

O	curvilinear
O	rotational
O	rectilinear
O	translation

Question # 5 of 10 (Start time: 03:54:22 PM, 13 June 2019) Total Marks: 1

The Stoke's theorem uses which of the following operation?

Select the correct option

O	Divergence
O	Gradient
O	Curl
O	Laplacian

Question # 6 of 10 (Start time: 03:32:38 PM, 13 June 2019) Total Marks: 1

Greens theorem applies only to _____ vector fields

Select the correct option

O	one dimensional
O	two dimensional
O	three dimensional
O	None of these

Total Marks: 1

The radial component of velocity for a particle moving in circular path is

Select the correct option

●	constant
○	radius itself
○	variable
○	zero

Click to Save Answer & Move to Next Question

Question # 3 of 10 (Start time: 03:42:50 PM, 13 June 2019) Total Marks: 1

The Stoke's theorem can be used to find which of the following?

Select the correct option

O	Surface area enclosed by a function in the given region
O	Volume enclosed by a function in the given region
O	Linear distance
O	Curl of the function

The Stoke's theorem uses which of the following operation?

○	Divergence
○	Gradient
○	Curl
○	Laplacian

Greens theorem applies only to _____ vector fields

○	one dimensional
○	two dimensional
○	three dimensional
○	None of these

The point, through which the whole weight of the body acts, irrespective of its position, is known as

○	Moment of inertia
○	Center of gravity
○	Center of percussion
○	Center of mass

Displacement of a body is a _____ quantity

- scalar
- vector
- scalar and vector
- none of these

Which of the following is NOT an application of Green's theorem?

O	Centroid of plane figures
O	Volume of plane figures
O	Area surveying
O	Solving two dimensional flow integrals

Which of the following is the simplest type of motion along a straight line path.

○	plane motion
○	rectilinear motion
○	curvilinear Motion
○	none of these

The normal component of acceleration is zero when motion of particle is

- O curvilinear
- O rotational
- O rectilinear
- O translation

According to Stokes' theorem if S is an open, two sided surface bounded by a simple closed curve C and

$\vec{A}$

has continuous first partial derivatives, then

Select the correct option

O	$\iint_S (\nabla \times \vec{A}) \cdot d\vec{S}$
O	$\iint_S (\nabla \cdot \vec{A}) \cdot d\vec{S}$
O	$\iint_S (\nabla \times \vec{A}) \, dS$
O	$\iint_S (\nabla \cdot \vec{A}) \, \vec{A} \, dS$

The Green's theorem can be related to which of the following theorems mathematically?

○	Gauss divergence theorem
○	Stoke's theorem
○	Euler's theorem
○	Leibnitz's theorem

When a particle moves along a straight path, then the particle has

- tangential acceleration only
- centripetal acceleration only
- both tangential and centripetal acceleration
- none of these

The Stoke's theorem uses which of the following operation?

Select the correct option

- Divergence
- Gradient
- Curl
- Laplacian

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- curvilinear
- rotational
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- translation

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○	Surface area enclosed by a function in the given region
●	Volume enclosed by a function in the given region
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○	Centroid of plane figures
○	Volume of plane figures
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The Green's theorem can be related to which of the following theorems mathematically?

- Gauss divergence theorem
- Stoke's theorem
- Euler's theorem
- Leibnitz's theorem

Which of the following theorem convert line integral to surface integral?

- Stoke's theorem only
- Gauss divergence and Stoke's theorem
- Green's theorem only
- Stoke's and Green's theorem

When a particle moves along a straight path, then the particle has

○	tangential acceleration only
○	centripetal acceleration only
○	both tangential and centripetal acceleration
○	none of these

Applications of Green's theorem are meant to be in

○	One dimensional
○	Two dimensional
○	Three dimensional
○	Four dimensional

Which of the following is the simplest type of motion along a straight line path.

○	plane motion
○	rectilinear motion
○	curvilinear Motion
○	none of these

The Stoke's theorem can be used to find which of the following?

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○	Volume enclosed by a function in the given region
○	Linear distance
○	Curl of the function

Question # 6 of 10 (Start time: 03:44:37 PM, 11 June 2019)

Both double and triple integrals can be used to calculate

Select the correct option

O	area under the curve
O	length of curve
O	volume of the given object
O	None of these

The normal component of acceleration is zero when motion of particle is

O	curvilinear
O	rotational
O	rectilinear
O	translation

The radial component of velocity for a particle moving in circular path is

●	constant
○	radius itself
○	variable
○	zero

Which of the following is NOT an application of Green's theorem?

○	Centroid of plane figures
○	Volume of plane figures
○	Area surveying
○	Solving two dimensional flow integrals

Displacement of a body is a _____ quantity

Select the correct option

- scalar
- vector
- scalar and vector
- none of these

Greens theorem applies only to _____ vector fields

O	one dimensional
O	two dimensional
O	three dimensional
O	None of these

The radial component of velocity for a particle moving in circular path is

○	constant
○	radius itself
○	variable
○	zero

The point, through which the whole weight of the body acts, irrespective of its position, is known as

○	Moment of inertia
○	Center of gravity
○	Center of percussion
○	Center of mass

The rate of change of tangential velocity of an object traveling in a circular orbit or path is known as

○	Normal acceleration
○	Tangential acceleration
○	Rectangular acceleration
○	None of these

According to Gradient Theorem if $\vec{A}$ is an open two sided surface bounded by a simple closed curve C and

$\vec{A}$

has continuous first partial derivatives then

Select the correct option

O	$\iint_{\cdot}$
O	$\iint_{\cdot}$
O	$\iint_{\cdot}$
O	$\iint_{\cdot}$

Question # 7 of 10 (Start time: 09:04:31 PM, 12 June 2019)

Which of the following is NOT an application of Green's theorem?

Select the correct option

○	Centroid of plane figures
●	Volume of plane figures
○	Area surveying

The Green's theorem can be related to which of the following theorems mathematically?

○	Gauss divergence theorem
○	Stoke's theorem
○	Euler's theorem
○	Leibnitz's theorem

[redacted]

[redacted]

[redacted]

[redacted]

○	$\nabla \cdot (\rho \mathbf{v}) + \frac{\partial \rho}{\partial t} = 0$
○	$\frac{\partial \rho}{\partial t} = 0$
○	$\nabla \cdot (\rho \mathbf{v}) = 0$
●	$\nabla \cdot (\rho \mathbf{v}) + \frac{\partial \rho}{\partial t} = 0$

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Classical mechanics deals with the

○	Statics objects only
○	microscopic objects only
○	macroscopic objects only
○	None of these

MTH622 Quiz#1 MTH622

Question # 3 of 10 (Start time: 03:02:43 PM, 12 June 2019)

Displacement of a body is a _____ quantity

Select the correct option

- scalar
- vector

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Question # 5 of 10 (Start time: 09:03:12 PM, 12 June 2019)

Both double and triple integrals can be used to calculate

Select the correct option

○	area under the curve
○	length of curve
○	volume of the given object

$$y = (4-x)^{1/2}$$

$$\frac{dy}{dx} = \frac{1}{2}(4-x)^{\frac{1}{2}-1} \cdot \frac{d}{dx}(4-x)$$

$$= \frac{1}{2}(4-x)^{-\frac{1}{2}} \cdot (-1)$$

$$= -\frac{1}{2(4-x)^{1/2}}$$

$$= -\frac{1}{2\sqrt{4-x}}$$

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

Math 622 25/6/18

* State Green's Theorem

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* Normalize $3i+2j+3k=4$

* If $A = xy^2i - 2xyj + zyk$ then find curl of A

* If mass and momentum given that Force field find kaun thi

* A ki value given thi and divergence theorem

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Mcqs easy thy tough ni thy
1. Divergence theorem ka question
2. Stocks theorem ka question tha
3. Irrotational proof tha 1 vector ko
4. Newton 2nd law of motion
Baqi 2 question me proof thy 2 choty choty

Mth622
Objectives conceptual
1.newton second law of motion
2.del. A=O AND del*A=0 ... Justify
2.A was given.. Find del. A
3.FOrce find mas and acceleration givn thi.
4.line integral find
5.Stoks law

Mth622
Surface integrl k 2 que
Ik directional derivative

4. Newton 2nd law of motion
Baqi 2 question me proof thy 2 choty choty

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Objectives conceptual
1.newton second law of motion
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Justify
2.A was given.. Find del. A
3.FOrce find mas and acceleration givn thi.
4.line integral find
5.Stoks law

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Surface integrl k 2 que
Ik directional derivative
Ik koi normal surface wala tha
Volume integralk long que tha
Ik green theroem sa que tha
Write statemt of green theroem

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(KEEP REMEMBER IN YOUR PRAYERS)

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